

Analysis First-Year Exam, May 2017, Part 2

Answer *FOUR* questions in detail. State carefully any results used without proof.

1. Let $(f_n)_{n=0}^\infty$ be a sequence of differentiable functions. Decide whether each implication is valid, giving proof or counterexample as appropriate:
 - (i) if $f_n \rightarrow f$ uniformly on $(-\infty, \infty)$ then $f_n^2 \rightarrow f^2$ uniformly on $(-\infty, \infty)$;
 - (ii) if $f_n \rightarrow f$ uniformly on $[-1, 1]$ then $\int_{-1}^1 f_n \rightarrow \int_{-1}^1 f$;
 - (iii) if $f_n \rightarrow f$ uniformly on $(-1, 1)$ then f is differentiable and $f'_n \rightarrow f'$ uniformly on $(-1, 1)$.

2. Let f be a continuous real-valued function on $[0, 1]$ and let $\alpha \in (0, \infty)$. Assume that

$$\int_0^1 t^{n\alpha} f(t) dt = 0$$

for all but finitely many values of $n \in \mathbb{N}$. What conclusions can be drawn about f ?

3. Let $\mathcal{F}$ be an equicontinuous family of real-valued functions on the compact metric space X . Denote by $A \subseteq X$ the set whose elements are precisely those $a \in X$ at which $\mathcal{F}$ is bounded in the sense that $\{f(a) : f \in \mathcal{F}\} \subseteq \mathbb{R}$ is bounded. Prove that A is both open and closed.
4. Let $(f_n)_{n=0}^\infty$ be a sequence of measurable real-valued functions. Decide whether each of the following sets is measurable:
 - (i) $\{\omega : (f_n(\omega))_{n=0}^\infty \text{ is unbounded}\}$;
 - (ii) $\{\omega : (f_n(\omega))_{n=0}^\infty \text{ is periodic}\}$;
 - (iii) $\{\omega : (f_n(\omega))_{n=0}^\infty \text{ has distinct terms}\}$.
5. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space that is finite in the sense that $\mu(\Omega) < \infty$. Let $(f_n)_{n=0}^\infty$ be a sequence of non-negative measurable functions converging pointwise to f on Ω . True or false (proof or counterexample):

$$\int_{\Omega} \frac{1}{1 + f_n} d\mu \rightarrow \int_{\Omega} \frac{1}{1 + f} d\mu \quad \text{as } n \rightarrow \infty.$$